



The Kidney in Obesity: Current Evidence, Perspectives and Controversies

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Abstract

Purpose of Review As obesity and chronic kidney disease (CKD) remain a public health issue, we aim to elaborate on their complex relationship regarding pathogenetic mechanisms and therapeutic potential as well. The purpose of this review is to enhance our understanding of the interplay between obesity and CKD in order to timely diagnose and treat obesity-related CKD.

Recent findings Obesity and CKD pose significant intertwined challenges to global health, affecting a substantial portion of the population worldwide. Obesity is recognized as an independent risk factor, intricately contributing to CKD pathogenesis through mechanisms such as lipotoxicity, chronic inflammation, and insulin resistance. Recent evidence highlights additional factors including hemodynamic changes and intestinal dysbiosis that exacerbate kidney dysfunction in obese individuals, leading to histologic alterations known as obesity-related glomerulopathy (ORG).

Summary This narrative review synthesizes current knowledge on the prevalence, pathophysiology, clinical manifestations, and diagnostic strategies of obesity-related kidney disease. Furthermore, it explores mechanistic insights to delineate current therapeutic approaches, future directions for managing this condition and controversies. By elucidating the multifaceted interactions between obesity and kidney health, this review aims to inform clinical practice and stimulate further research to address this global health epidemic effectively.

Keywords Chronic kidney disease · Glomerular hyperfiltration · Glomerulopathy · Inflammation · Insulin resistance · Lipotoxicity · Melatonin · Microbiota · Obesity · RAAS · Tirzepatide

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Introduction

Obesity, as defined by the World Health Organization (WHO), is a condition where an individual's body mass index (BMI) exceeds 30 kg/m² [1]. This condition poses a significant global health challenge, currently impacting nearly 600 million adults worldwide [2]. Projections suggest that by 2030, approximately 20% of the global population over the age of five will have a BMI of 30 kg/m² or more [3]. Research indicates that obesity prevalence is higher in women compared to men across all age groups, while the age-standardized prevalence of obesity has surged from 4.6% in 1980 to 14.0% in 2019 [4].

Chronic kidney disease (CKD) is another major health concern, affecting over 800 million people globally, which constitutes more than 10% of the general population [5]. CKD is most prevalent among women, older adults, and individuals with diabetes mellitus or hypertension [6]. It is characterized by persistent kidney injury lasting at least three months, as indicated by glomerular filtration rate (GFR) values and proteinuria presence. CKD is classified into five stages based on GFR reduction and the albumin/creatinine ratio [7].

Recent studies have established obesity as an independent risk factor for CKD, irrespective of the presence of diabetes mellitus, arterial hypertension, or other known risk factors. In the United States, nearly half of the population suffers from both CKD and obesity [8]. By 2040, CKD is projected to become the fifth leading cause of death, partly due to the increasing prevalence of obesity [9]. Although the reported prevalence of obesity-related CKD varies, studies consistently agree on its rising incidence across all ethnic groups [10]. U.S. data from the National Health and Nutrition Examination Survey (NHANES) indicate that obese individuals have a significantly higher lifetime risk of CKD at all stages compared to those with normal weight (41.0% vs. 32.5%) [11]. A meta-analysis by Wang et al. found that obesity contributes to up to 33% of all renal disease cases in the United States. The same meta-analysis predicted that around 14% of males and 25% of females in industrialized countries develop CKD as a result of being overweight or obese [12].

Overall, the estimated incidence of obesity-related CKD is 2.7% while obesity constitutes a potential cause of around 15–30% of cases with CKD [13, 14]. A meta-analysis showed that around 14% and 25% of males and females respectively develop CKD as a clinical implication of excess body weight in industrialized countries [14]. Moreover, evidence indicates that overweight adults (i.e., those with a BMI > 25 kg/m²) have a five-fold higher risk of developing CKD compared to lean individuals. Furthermore, obesity itself is a significant predictor of end-stage

renal disease (ESRD) [15]. On the contrary, approximately 20–50% of patients with type 2 diabetes mellitus (T2DM) will eventually develop diabetic kidney disease (DKD). Globally, DKD is the primary cause of CKD and ESRD, accounting for 50% of all cases [16].

Evidence increasingly suggests that the association between obesity and CKD can begin as early as in maternal life. Studies have shown that factors such as low birth weight, reduced nephron number, preterm delivery, and intrauterine growth restriction are linked to higher risks of hypertension, obesity, albuminuria, and renal disease later in life [17, 18]. Obesity is more prevalent in individuals with a genetic predisposition, though the exact genetic contribution is unclear. Studies have shown a link between both central and general adiposity and CKD. In a UK Biobank study of 281,228 individuals, a 0.06 increase in waist-to-hip ratio correlated with a 30% higher CKD risk, while a 5-kg/m² higher BMI increased CKD risk by about 50% [19]. Genetic studies also suggest that diabetes, hypertension, and related factors explain most of the adiposity-CKD association, more so than traditional methods indicate [20]. Current data also indicate that abdominal obesity is associated with CKD. Higher waist circumference is related to albuminuria, decreased GFR, and the onset of end-stage renal disease (ESRD), independent of BMI [21]. Furthermore, higher BMI correlates with the presence and progression of proteinuria even in adults without kidney disease and is linked to the development of low estimated GFR and more rapid GFR decline over time. These findings hold true across diverse ethnic populations [22, 23].

Overall, obesity is a significant predisposing factor for the development and progression of CKD. As such, it should be targeted in public health initiatives aimed at reducing CKD prevalence. This review aims to consolidate current evidence regarding the underlying pathophysiological mechanisms, clinical presentation, and diagnostic approaches related to obesity-associated CKD. Additionally, it seeks to clarify treatment options and highlight future perspectives, and ongoing controversies in the literature on this topic.

Pathophysiology

Excessive fat accumulation in humans, seen commonly in obesity, can extend beyond fat tissue to organs like the kidneys, known as “lipotoxicity”. This condition disrupts lipid balance and cellular composition, leading to harmful lipid buildup [24]. According to the “Adipose Tissue Expandability Hypothesis”, fat tissue has a finite capacity to expand. When this limit is exceeded, excess lipids spill into the bloodstream and deposit in various organs [25]. In the renal context, lipotoxicity results in the accumulation of free fatty acids (FFAs), triglycerides, and ceramides, which contribute to kidney damage [26].

Renal lipotoxicity arises from multiple pathophysiological mechanisms such as inflammation, oxidative stress from reactive oxygen species (ROS), insulin resistance, and activation of the renin-angiotensin system [27]. Furthermore, the presence of apolipoprotein L1 (APOL1) gene polymorphisms may result in the inhibition of cholesterol efflux and cholesterol accumulation in the kidney, and particularly in the podocytes [28]. These mechanisms impair organelle function, disrupt intracellular signaling pathways, promote chronic inflammation, and increase susceptibility to cell death [27].

Chronic Inflammation

It is widely known that obesity is a state of chronic low-grade inflammation, characterized by increased infiltration of white adipose tissue (WAT) with macrophages. In particular, macrophages may constitute up to 40% of cells in WAT, while under normal circumstances they consist less than 10% of the cell's population [29]. Furthermore, macrophages are classified as M1 macrophages, which secrete pro-inflammatory cytokines and M2 macrophages, which release anti-inflammatory cytokines, the most well recognized being interleukin-4 (IL-4), IL-10 and interleukin-1 receptor antagonist (IL-1Ra). In obesity, there is a phenotypic switch from M2 to M1 macrophages, which produce a variety of pro-inflammatory cytokines, such as IL-6, IL-8, IL-1 β , IL-12, IL-23 and tumor necrosis factor alpha (TNF- α) [30]. Thus, this phenotypic switch from M2 to M1 macrophages creates an inflammatory milieu, accounting for an increased production of ROS and cellular damage in the kidney, among other target organs. Moreover, as only recently reported, macrophages exhibiting high expression of triggering receptor expressed on myeloid cells 2 (TREM2) play a pivotal role in obesity and diabetes related kidney injury. This novel finding that TREM2^{high} macrophages may possess beneficial properties regarding obesity and diabetes kidney disease could expand our knowledge and therapeutic potential regarding inflammation in obesity-related kidney injury [31].

Apart from macrophages, adipocytes in WAT also secrete adipocytokines, i.e. adipocyte-derived hormones that play a crucial role in obesity. More specifically, leptin, visfatin and resistin exhibit inflammatory and fibrotic properties, whereas adiponectin exerts anti-inflammatory and anti-fibrotic potential [14]. Among patients with obesity and CKD, adipocytes in WAT produce large amounts of leptin, visfatin and resistin, whereas adiponectin's production is reduced. Increased levels of leptin have been associated with inflammation, enhancement of ROS as well as accumulation of collagen and stimulation of transforming growth factor 1 β (TGF-1 β). In addition, increased levels of resistin have been related to the up-regulation of IL-1 and TNF- α

together with an aggregation of adhesion molecules, such as vascular cell adhesion molecule-1 (VCAM-1) and intercellular adhesion molecule-1 (ICAM-1) in the kidney. Besides, visfatin or nicotinamide phosphoribosyltransferase has been correlated with increased VCAM-1, IL-6 and TNF- α levels as well as enhanced production of extracellular matrix (ECM) [32]. Moreover, visfatin has been suggested to exert its inflammatory properties via activation of the inhibitory kappa B kinase/nuclear factor kappa B (IKK/NF- κ B) and the janus kinase 2/signal transducer and activator of transcription 3 (JAK2/STAT3) molecular pathways [33, 34]. Taken together, there is accumulating data supporting the role of the adipocytokines leptin, resistin and visfatin in inflammation together with fibrosis and progression to CKD. In sharp contrast, adiponectin, which is the most abundant adipocytokine under normal circumstances, is reduced in obesity.

Adiponectin is the major adipocytokine, which possesses anti-inflammatory features, that are mainly attributed to inhibition of NF- κ B and polarization of macrophages from M2 to M1. This dual action results in a decrease in the production of pro-inflammatory cytokines. Adiponectin acts by activating two receptors: adiponectin receptor-1 (ADIPOR1) and adiponectin receptor-2 (ADIPOR2), while its action may be modified by T-cadherin. Adiponectin exerts its protective effects on the kidney by binding to the predominant receptor in proximal tubular and glomerular cells ADIPOR1 [35]. However, we should not overlook the fact that serum adiponectin levels may be increased in patients with obesity and CKD due to its decreased clearance by the kidneys [36, 37]. Notably, serum adiponectin concentration has been documented to be increased, especially among patients with CKD stage 5 [37].

Other significant inflammatory molecules secreted from adipocytes in WAT are IL-6, TNF- α and Plasminogen Activator Inhibitor-1 (PAI-1). PAI-1 or SERPINE1 is the normal inhibitor of tissue plasminogen activators (tPAs) and is strongly associated with thrombus formation in the context of inflammation [30]. It is noteworthy that in obesity, adipocytes in WAT enlarge from 30 μ m to more than 100 μ m, thus becoming hypertrophic [29]. This remarkable hypertrophy creates a hypoxic milieu driven by the increased production of hypoxia-inducible factor-1 α (HIF-1 α). In turn, HIF-1 α promotes the release of pro-inflammatory cytokines, such as IL-6 and TNF- α as well as ROS, thereby enhancing release of PAI-1 [30, 38]. Furthermore, chemokines, such as monocytes chemoattractant protein-1 (MCP-1), MCP-2, MCP-3 and MCP-4, eotaxin, chemokine motif ligand-5 (CCL-5) and chemokine CXC motif ligand-10 (CXCL-10) are also stimulated in obesity [30].

Recently, the role of the nucleotide-binding oligomerization domain like receptor protein 3 (NLRP3) inflammasome in obesity-related nephropathy is being extensively studied.

In particular, NLRP3 has been demonstrated to be regulated by acid sphingomyelinase (ASM) in podocytes in an animal model of obesity [39]. ASM resulted in the regulation of activation of the NLRP3, increased production of IL-1 β /IL-18 and glomerular inflammation and injury [39]. As a multiprotein signaling complex, NLRP3 inflammasome, when activated, results in a caspase-1 mediated enhancement in the release of the pro-inflammatory cytokines IL-1 β /IL-18. Thus, its activation promotes inflammatory changes together with cellular damage in the kidney [40]. Figure 1 highlights the key molecular inflammatory mechanisms by which obesity predisposes individuals to CKD.

Oxidative Stress

Reactive oxygen species play a pivotal role in the detrimental effects of fat accumulation in the kidneys of individuals with obesity. They affect intracellular organelles like the endoplasmic reticulum (ER) and mitochondria, influencing autophagy and apoptosis. The ER is vital for maintaining cellular homeostasis by managing protein synthesis, lipid

metabolism, and calcium regulation [41]. Lipid accumulation and oxidative stress can disrupt these functions, triggering the unfolded protein response (UPR). Prolonged activation of the UPR leads to ER stress, which can induce apoptosis through pathways such as protein kinase RNA like ER kinase (PERK), inositol requiring enzyme 1 (IRE1), and activating transcription factor 6 (ATF6) [42]. In the kidneys, free fatty acids, like palmitic acid, activate these pathways in specific cell types, highlighting diverse cellular responses to lipid-induced ER stress [42]. The relationship between ER function and oxidative stress is complex. The ER produces ROS during protein folding, while ROS can induce ER stress, activating the UPR and potentially causing cell damage. Studies indicate that lipid overload, particularly from PA, exacerbates ER stress via oxidative pathways, creating a feedback loop that worsens renal damage [43, 44].

Mitochondria are essential for kidney energy metabolism. High-fat diets can alter mitochondrial morphology and disrupt biogenesis, dynamics, and mitophagy. Oxidative stress influences these disruptions, leading to mitochondrial dysfunction [45, 46]. ROS further impact mitochondrial

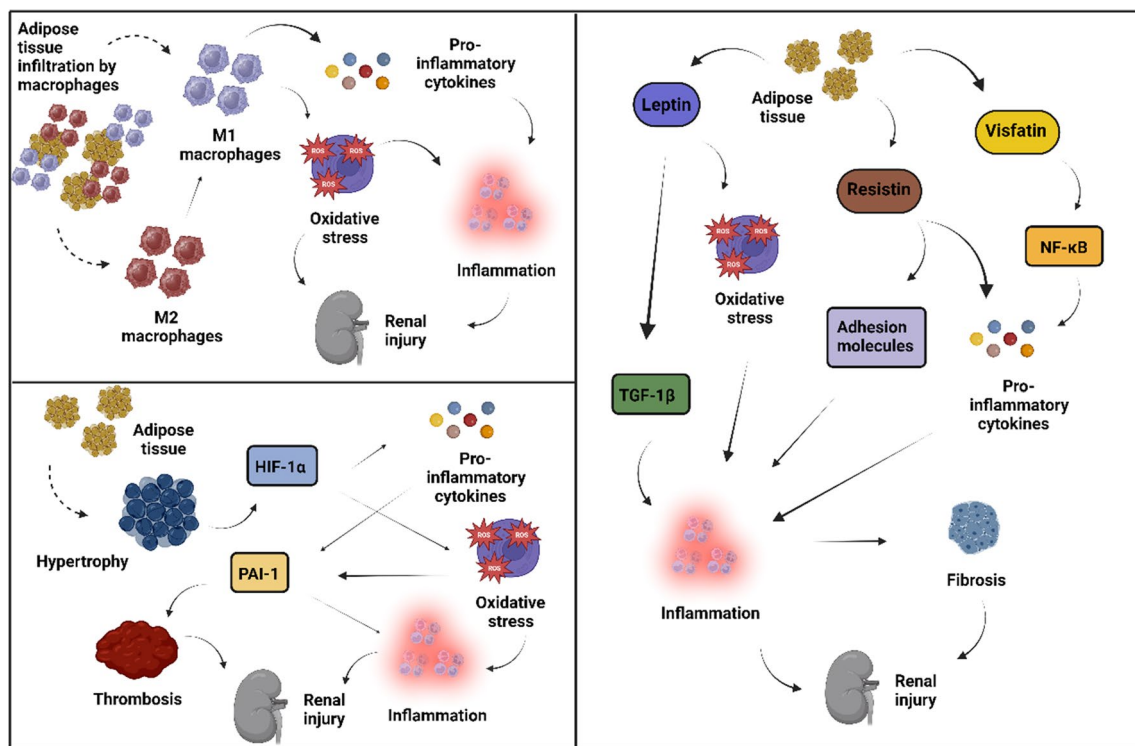


Fig. 1 Presents the molecular inflammatory pathways by which obesity predisposes individuals to chronic kidney disease (CKD). The figure highlights the role of WAT macrophage infiltration, particularly the shift from anti-inflammatory M2 to pro-inflammatory M1 macrophages. This shift contributes to an increased production of ROS and cellular damage. Additionally, adipocytokines like leptin, visfatin, and resistin promote inflammation and fibrosis. The figure also

depicts hypoxia induced by adipocyte hypertrophy, emphasizing the role of HIF-1 α in promoting inflammation and thrombosis, thereby contributing to the progression of obesity-related CKD. Abbreviations: HIF-1 α : Hypoxia Induced Factor 1 α ; NF- κ B: Nuclear Factor kappa-light-chain-enhancer of Activated B cells; PAI-1: Plasminogen Activator Inhibitor-1; TGF-1 β : Transforming Growth Factor 1 β . Created with www.BioRender.com. (assessed on 24 July 2024)

integrity, promoting fragmentation and reducing biogenesis, as shown by the decreased expression of genes like ATP5a-1, PGC-1 α , and NDUFS8, indicating a cycle of dysfunction in lipotoxic kidney injury [41, 47]. Peroxisome proliferator activated receptor (PPAR) pathways play a role in kidney impairment via oxidative stress. PPAR- α modulates oxidative stress through pathways such as AMPK/AKT/GSK3 β /Fyn and PI3K/AKT/FOXO3a [48]. PPAR- γ is also crucial in obesity-related kidney disease, regulating glucose and lipid metabolism. Mutations in PPAR- γ can lead to dysfunctional lipid homeostasis, contributing to renal damage. In animal models, PPAR- γ deficiency is linked to altered renal lipid metabolism and increased susceptibility to renal injury, whereas low levels of adiponectin, regulated by PPAR- γ , are associated with albuminuria in obese subjects [49–51].

Autophagy, essential for clearing damaged proteins and organelles, plays a protective role in kidney health. While oxidative stress can initially inhibit excessive autophagy, over time, it may impair autophagic flux, leading to increased ROS [52]. Specifically, palmitic acid can initially activate autophagy but may disrupt autophagic flux over time, likely due to lysosomal dysfunction, indicating a complex interplay with lipid metabolism [53]. Autophagy, in turn, helps reduce oxidative stress by clearing ROS and excess lipids. Deficiency in autophagy can lead to increased ROS, highlighting its protective role. This dynamic interplay evolves throughout kidney disease progression, with autophagy mitigating oxidative stress by reducing excess ROS and lipids [54].

Insulin Resistance

Insulin resistance is defined as the reduced effectiveness of insulin in its target organs (liver, skeletal muscles, adipose tissue), often manifesting as glucose intolerance, hyperglycemia, and hyperinsulinemia. The kidney plays a crucial role in glucose homeostasis, and renal dysfunction in CKD may disrupt glucose regulation, contributing to insulin resistance [55]. This is particularly common in obesity-related CKD, where several factors exacerbate the condition.

In obese individuals, triglyceride accumulation leads to adipocyte hypertrophy, which results in the secretion of pro-inflammatory cytokines, free fatty acids (FFAs), and leptin, collectively promoting insulin resistance [56]. As mentioned above, inflammation plays a central role in the development of insulin resistance in obesity-related CKD. Impaired adipogenesis leads to adipocyte hypertrophy and the recruitment of adipose tissue macrophages (ATMs). Both hypertrophic adipocytes and ATMs produce pro-inflammatory cytokines [30–33].

Experimental studies have shown that these cytokines are pivotal in the onset and progression of insulin resistance. For instance, TNF- α produced by hypertrophic adipocytes

impairs insulin activity by increasing the phosphorylation of insulin receptor substrate-1 (IRS-1) [57]. Conversely, the absence of TNF- α is associated with lower circulating FFAs, providing protection against insulin resistance [58]. FFAs, significant contributors to lipotoxicity, accumulate as intracellular and intrahepatic lipids, inhibiting insulin action in target organs. Once inside cells, FFAs convert to diacylglycerol (DAG), activating protein kinase C (PKC) and c-Jun N-terminal kinase (JNK), which promote the phosphorylation of IRS-1, downregulating insulin signaling [59].

Additionally, elevated leptin levels in obese individuals contribute to insulin resistance. Although leptin typically regulates appetite and energy expenditure, high levels in obesity lead to leptin resistance, paradoxically promoting insulin resistance. This is due to leptin's role in upregulating pro-inflammatory cytokines, further enhancing inflammation and insulin resistance [60, 61].

The Renin–Angiotensin–Aldosterone System (RAAS)

The renin–angiotensin–aldosterone system (RAAS) is a fundamental peptidergic and hormonal network crucial for maintaining blood pressure, electrolyte balance, and fluid homeostasis. Originating primarily in the kidneys, RAAS responds to decreased renal blood flow by activating renin, which cleaves angiotensinogen to form angiotensin I. Subsequent conversion by angiotensin-converting enzyme (ACE) yields angiotensin II, a robust vasoconstrictor and stimulator of aldosterone release from the adrenal cortex. Aldosterone promotes sodium retention in the kidneys, thereby influencing fluid volume and blood pressure regulation [62, 63].

Beyond its systemic effects, RAAS exerts local influences within tissues, including visceral adipose tissue, where its dysregulation is implicated in obesity-related complications. This local activity can influence local tissue functions independently of its systemic effects, although interactions between systemic and tissue-specific RAAS components are recognized [64]. In obese individuals, despite an anticipated state of salt and volume retention suppressing renin secretion, systemic and local RAAS activation persists. This aberrant activation is fostered by the inflammatory milieu of adipose tissue, exacerbating systemic hypertension and metabolic dysfunction [65]. Furthermore, obesity is linked to heightened activation of the sympathetic nervous system and modulation of the gastro-renal axis [66]. These factors collectively contribute to systemic and/or intraglomerular hypertension. The simultaneous activation of the RAAS and the sympathetic nervous system disrupts natriuresis and increases renal tubular salt reabsorption. This results in volume expansion and the development of hypertension [67]. Moreover, obesity contributes to RAAS activation through increased adipose tissue mass, which can lead to elevated

intra-abdominal pressure and potential kidney compression, further exacerbating hypertension [68].

Adipocytes contribute significantly to RAAS activity by producing angiotensinogen, a precursor of angiotensin peptides. Experimental studies have shown that removing adipose-derived angiotensinogen can prevent the development of obesity-related hypertension [69]. Elevated angiotensin II levels in obesity promote renal injury by increasing intraglomerular pressure, fostering proteinuria, and stimulating pro-inflammatory and fibrotic pathways in renal cells [70]. Notably, angiotensin II induction of epithelial-mesenchymal transition (EMT) and its interaction with matrix metalloproteinases (MMPs) underscore its role in renal fibrosis progression [71]. Aldosterone, another key RAAS effector, also emerges from sources beyond adrenal glands, including adipose tissue. Elevated aldosterone levels in obesity contribute to insulin resistance and sodium retention, exacerbating metabolic dysregulation and renal fibrosis through activation of mineralocorticoid receptors in adipocytes. This activation augments adipogenesis and

inflammation, while stimulating profibrotic factors such as TGF- β and PAI-1, crucial for extracellular matrix deposition in the kidney [72, 73].

The intricate interplay of RAAS components in obesity highlights their role not only in systemic hypertension but also in the pathogenesis of CKD. Strategies targeting RAAS components have shown promise in mitigating renal fibrosis and halting disease progression, suggesting therapeutic perspectives to address obesity-related nephropathies. Figure 2 provides a schematic illustration of how obesity contributes to the development and progression of CKD through the activation of the RAAS system.

Glomerular Hyperfiltration

The GFR serves as a critical indicator of kidney function, with elevated GFR levels often associated with glomerular hyperfiltration (GH). GH plays a pivotal role in the progression of kidney diseases, particularly diabetic nephropathy, where it signifies an early stage of CKD

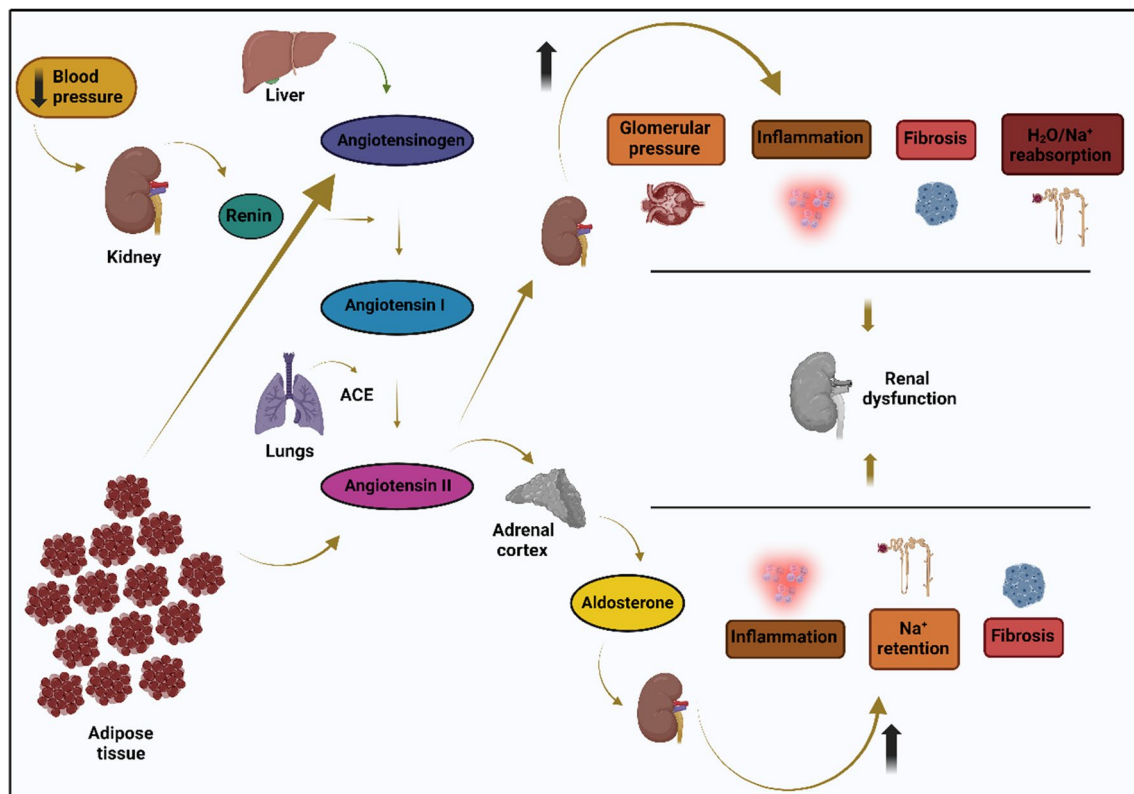


Fig. 2 Illustrates the complex interactions within the renin–angiotensin–aldosterone system (RAAS) in the context of obesity. This schematic highlights how obesity-induced RAAS activation contributes to chronic kidney disease (CKD) through mechanisms such as systemic and intraglomerular hypertension, increased sodium retention, and renal fibrosis. The figure underscores the role of adipose tissue in

producing RAAS components, exacerbating hypertension and metabolic dysfunction, and promoting renal injury via inflammatory and fibrotic pathways. Finally, it emphasizes the potential therapeutic targets within the RAAS for addressing obesity-related nephropathies. Abbreviations: ACE: Angiotensin Converting Enzyme. Created with www.BioRender.com. (assessed on 24 July 2024)

[74]. Additionally, GH is prevalent among individuals affected by obesity, potentially establishing a direct link between obesity and renal disease. Left untreated, GH can lead to gradual deterioration in kidney function [75]. Histologically, GH manifests as enlargement of both glomerular and tubular structures, primarily due to initial changes in intrarenal hemodynamics. These changes stem from alterations in the tone of afferent and efferent arterioles, resulting in heightened renal blood flow and increased intraglomerular pressure. Such hemodynamic shifts are influenced by various factors including inflammation, oxidative stress, hyperinsulinemia, activation of the RAAS, and enhanced sodium absorption [76, 77].

A study of 342 obese individuals, with an average BMI of 47.6 kg/m² and no primary kidney disease, found that GFR increases with body weight and correlates with albuminuria, indicating early kidney damage [78]. Detecting albuminuria early is vital, as untreated albuminuria can progress to glomerulosclerosis and nephropathy. Research on non-diabetic African Americans shows that obesity is linked to higher GFR values, effective renal plasma flow (ERPF), filtration fraction, and a high prevalence of GH. GH can elevate proximal tubular flow and reabsorption of small- and large-molecule solutes, leading to glomerular enlargement, tubulointerstitial inflammation, and fibrosis [79].

Additionally, GH may cause mechanical stress on the filtration barrier and subsequent structures, resulting in podocyte loss and focal segmental glomerulosclerosis (FSGS) [80]. A cross-sectional study of 151 metabolically healthy obese (MHO) individuals aged 40 and older found that MHO is associated with a higher risk of hyperfiltration and mildly reduced estimated GFR, as well as early kidney dysfunction. Serum uric acid partially mediates this association, underscoring the complex metabolic interactions in GH and kidney health [81].

Grasping the mechanisms and consequences of GH is vital for developing strategies to prevent and manage kidney disease, especially in those at risk due to obesity and metabolic disorders. Early intervention and targeted therapies to control GH and its contributing factors could significantly help in preserving renal function and preventing CKD progression.

Microbiota

The gut microbiota comprises the sum of bacteria, viruses, fungi and archaea that inhabits the human intestines. Under normal circumstances, there is homeostasis between the human gut microbiota and the host. However, under abnormal conditions, this state of homeostasis is disrupted resulting in an imbalance between the gut microbiota and the host, widely known as “gut dysbiosis” [82]. Nowadays, obesity as a typical

example of gut dysbiosis is at the forefront of research. It is characterized by an increased ratio of the Gram positive Firmicutes to the Gram-negative Bacteroidetes (F/B ratio) [83, 84]. Despite the fact that most animal studies advocate the increased F/B ratio as a hallmark of obesity, studies in humans are inconclusive [85, 86]. These discrepancies may be attributed to differences in molecular methods, such as polymerase chain reaction (PCR) or next generation sequencing (NGS) used to describe the human gut microbiota as well as differences in the studied populations.

Nevertheless, there is mounting evidence supporting the notion that dysbiosis in obesity is mainly characterized by reductions in the diversity and the composition of the gut microbiota [87–89]. These alterations in the gut microbiota may lead to several changes regarding the production and synthesis of various bacterial metabolites. Increased trimethylamine (TMA) and branched chain fatty acids (BCFAs) production from the gut microbiota and decreased production of short chain fatty acids (SCFAs) have detrimental effects on lipids metabolism and inflammation. Apart from the involvement of various bacterial metabolites, the dysregulation of tight junctions (TJs) in the intestinal epithelial cells results in the so-called “leaky gut”. In turn, “leaky gut” permits the invasion of the lipopolysaccharide (LPS) of Gram-negative bacteria in the systemic circulation resulting in increased serum LPS levels and systemic inflammation [90]. Furthermore, as kidney dysfunction evolves, uremic toxins, such as p-cresyl and indoxyl accumulate in the gut and are then conjugated to p-cresyl sulfate (PCS) and indoxyl sulfate (IS) in the liver [91].

Overall, this intricate interplay between obesity, progression to CKD and the gut microbiota has been suggested to be mediated via the “leaky gut”. This disruption in TJs allows for the development of an inflammatory milieu in the gut and the kidneys as well. Figure 3 illustrates the primary pathophysiological mechanisms involved in obesity-related CKD.

Clinical Presentation

Obesity-related CKD often develops gradually, characterized by proteinuria, and occasionally by microscopic hematuria or renal impairment. This condition frequently presents without histological or clinical evidence of other renal disorders. The diagnosis of obesity-related kidney disease relies on the following clinical and histological criteria: 1) A BMI ≥ 30 kg/m² is typically required for diagnosis, along with a waist circumference exceeding 90 cm for men and 85 cm for women, respectively. 2) The presence of proteinuria at various levels, generally exceeding 0.3 g per 24 h, although severe proteinuria and edema are infrequent. 3) Renal function is usually normal or only mildly abnormal, whereas microscopic hematuria may be

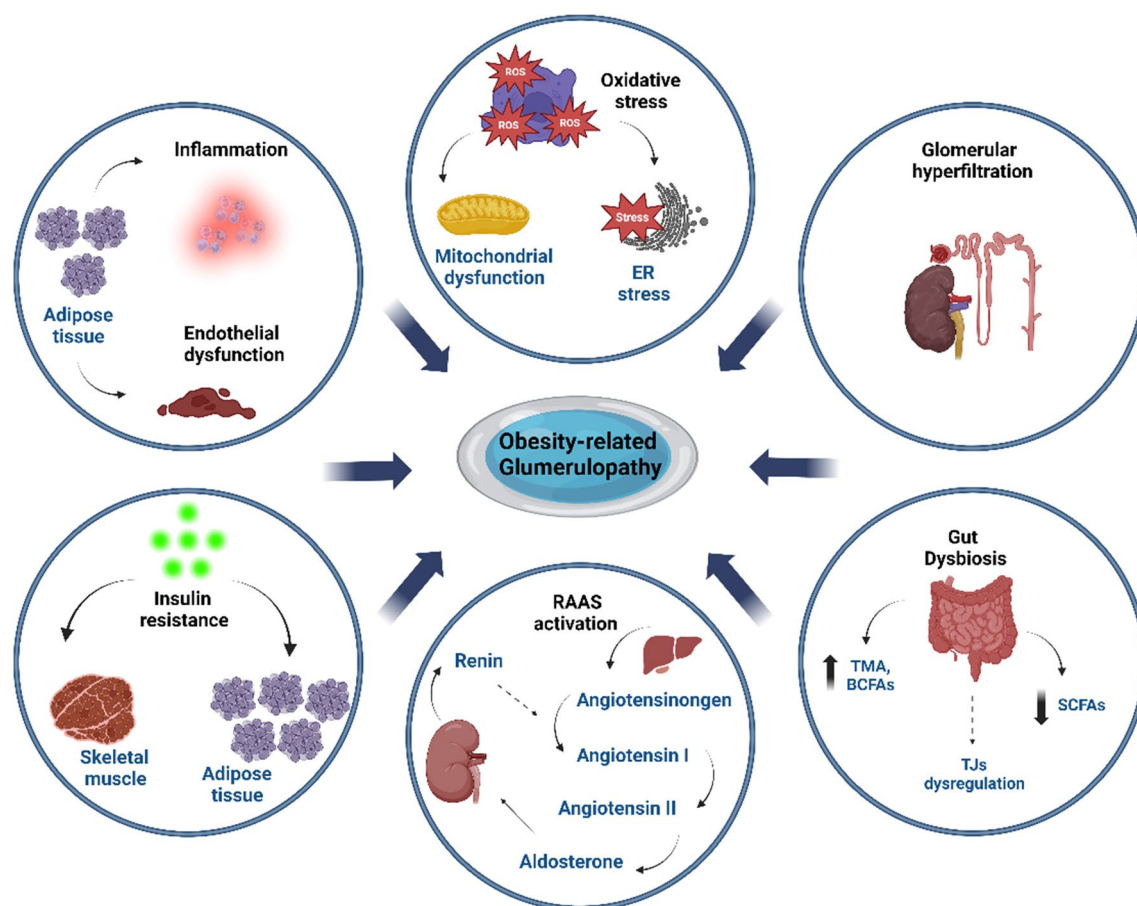


Fig. 3 Depicts a schematic presentation of the primary pathophysiologic mechanisms behind obesity-related CKD. Lipotoxicity results in inflammation, oxidative stress, lipid accumulation, insulin resistance, altered hemodynamics, and gut dysbiosis, all contributing to renal damage. The interplay of these factors exacerbates kidney function decline in individuals with obesity. Abbreviations: BCFAs: Branched

Chain Fatty Acids; ER: Endoplasmic Reticulum; RAAS: Renin–Angiotensin–Aldosterone System; ROS: Reactive Oxygen Species; SCFAs: Short Chain Fatty Acids; TJs: Tight Junctions; TMA: Trimethylamine. Created with www.BioRender.com. (assessed on 24 July 2024)

observed. 4) A significant increase in glomerular volume is often observed, with light microscopy sometimes revealing FSGS. 5) Electron microscopy typically shows fusion of epithelial cell foot processes. 6) Other primary or secondary glomerulopathies must be ruled out to confirm the diagnosis [92]. Table 1 depicts main studies associating BMI and WC with renal function outcomes during the past ten years.

Diagnostic Evaluation

The diagnosis of obesity-related CKD utilizes biomarkers and imaging techniques common in general CKD assessments, with kidney biopsy providing additional

valuable insights. Biomarkers help evaluate kidney function and detect damage, while imaging can assess structural changes. A biopsy may reveal specific pathophysiological changes linked to obesity, offering a more detailed understanding of the disease's impact.

Biomarkers

For obesity-related CKD, there are no current universal screening standards. However, the same screening tests for diabetic CKD could be applied such as testing of urine microalbumin/creatinine ratio annually and eGFR yearly [13, 104]. The primary biomarkers include the GFR and renal plasma flow (RPF) [105]. In clinical practice, GFR is usually estimated using equations based on plasma levels of serum creatinine (SCr) and cystatin C (sCysC), or a combination of both [106]. Diagnosing CKD in obese individuals

Table 1 List of main studies relating BMI and WC with renal function outcomes in the past ten years

Author/Year	Study Population	Findings	Remarks
Yang et al., 2024 [93]	130,504 Korean participants aged ≥ 60 y.o	BMI/WC was not related to a decreased eGFR	The presence of MS as well as the number of MS components were related to a decreased eGFR
Cejka et al., 2023 [94]	4,537 German patients with mild to severe CKD aged ≥ 18 y.o	WC but not BMI was associated with increased risk of death from any cause	NC in women but not in men was also associated with increased risk of death from any cause
Bullen et al., 2022 [95]	27,550 African-American participants in the Southeast USA aged ≥ 45 y.o	BMI/WC were associated with decline in eGFR	VAI/LAP were related to higher odds of decline in eGFR
Gao et al., 2021 [96]	10,251 patients with T2DM in the USA and Canada aged ≥ 18 y.o	BMI/WC and FMI were equally predictive of a decline in eGFR	FMI was calculated as FM in kilograms by Lee et al. in the NHANES 1999–2006 divided by height in meters square
Cai et al., 2021 [97]	7,593 participants from Shanghai, China aged 12–93 y.o	WC predicted a decline in eGFR over time	Combined BMI/WC predicted moderate CKD
Moreira et al., 2021 [98]	14,636 participants from Elsa-Brazil aged ≥ 18 y.o	BMI/WC were associated with CKD over time	BMI/WC were related to decline in eGFR and development of proteinuria as well
Cho et al., 2020 [99]	514,866 Korean participants aged ≥ 18 y.o	Metabolically healthy individuals with obesity according to BMI showed improvement in eGFR when changing to metabolically healthy individuals without obesity	Metabolically healthy individuals with obesity according to WC also showed improvement in eGFR when converted to metabolically healthy individuals without obesity
Yang et al., 2019 [100]	4,221 participants from rural China aged 27–95 y.o	BMI/WC/WHtR were associated with development of CKD	WHtR was a better predictor of CKD development, in particular among female individuals
Madero et al., 2017 [101]	2,489 individuals in the USA aged between 70–79 y.o	BMI/WC were associated with decline in eGFR	VAF was a better predictor of incident CKD than BMI/WC
Davis et al., 2016 [102]	903 Australian patients with CKD stage 3–4, aged > 18 y.o	Increased BMI is related to decreased risk of progression of CKD stage 3–4 and all-cause mortality	Unlike BMI, WC was not associated with any prediction regarding progression of CKD stage 3–4 and all-cause mortality
Oh et al., 2014 [103]	390 Korean participants aged > 65 y.o	BMI/WC was not associated with a decline in eGFR	However, BFP was associated with a reduction in eGFR

Abbreviations: BFP: Body Fat Percentage; BMI: Body Mass Index; CKD: Chronic Kidney Disease; eGFR: estimated Glomerular Filtration Rate; FM: Fat Mass; FMI: Fat Mass Index; LAP: Lipid Accumulation Product; MS: Metabolic Syndrome; NC: Neck Circumference; VAI: Visceral Adiposity Index; VAT: Visceral Abdominal Fat; WC: Waist Circumference; WHtR: Waist to Height Ratio; y.o.: years old

presents unique challenges due to the influence of obesity on various biomarkers. SCr levels are affected by factors such as muscle mass, growth, diet, and physical activity, making SCr a less reliable marker in obese patients [107]. Additionally, the lack of standardization in SCr assays complicates its reliability further [108]. Therefore, creatinine-based formulas often incorporate surrogates for muscle mass, such as height, weight, and gender, to enhance accuracy. Cystatin C, a non-glycosylated protein produced at a constant rate by most nucleated cells and filtered by the glomerulus, offers an alternative [109]. While GFR is the primary determinant of cystatin C concentration, it can still be influenced by various conditions, including obesity, though to a lesser extent than SCr [110].

To overcome these limitations, recent advancements have introduced various urinary biomarkers, such as urinary cystatin C, N-acetyl- β -D-glycosaminidase, and urinary neutrophil gelatinase-associated lipid deposition protein. These biomarkers provide additional tools for assessing renal function and detecting early kidney damage. Imaging modalities also play a crucial role in evaluating the structural and functional aspects of the kidneys in obese patients [111, 112].

Recent research has highlighted the potential of novel biomarkers and molecular insights in diagnosing obesity-related CKD. MicroRNAs (miRNAs) play a significant role in the progression of kidney fibrosis, a common pathway leading to ESRD [113, 114]. Specific miRNAs, such as miR-21 and the miR-29 family, are involved in obesity-related inflammation and fibrosis pathways. For instance, miR-21 is associated with obesity-related inflammation, and its inhibition has been linked to reduced body weight and adipocyte size [115]. Similarly, miR-29c has been identified as a marker for maternal obesity, influencing pre- and postnatal growth through inflammation pathways [116].

Imaging Techniques

Renal ultrasonography is a noninvasive, simple, and safe imaging test, primarily used to assess kidney size, morphological structure, parenchymal thickness, and echogenic intensity. Its repeatability makes it ideal for ongoing monitoring of kidney health [117]. Computed tomography (CT) offers a precise, high-quality examination of kidney structure and function, allowing for the quantification of ectopic adipose tissue associated with obesity-related CKD. CT imaging measures adipose tissue density in Hounsfield units, providing detailed insights into fat accumulation in the kidneys [118]. Magnetic resonance imaging (MRI) is another valuable tool for assessing ectopic lipid buildup in the kidneys. It offers high-resolution images that can help identify early changes in kidney morphology and function [119].

Kidney Biopsy

Kidney biopsy provides valuable diagnostic insights into obesity-related glomerulopathy (ORG) and is generally considered safe [120]. In 2001, Kambham et al. conducted the first large-scale study analyzing 6818 native renal biopsies, which revealed a 10-fold increase in ORG incidence over 15 years. This study found that patients with ORG were typically older, predominantly Caucasian, had a lower occurrence of nephrotic syndrome, and showed less than 50 percent foot process effacement on electron microscopy. The BMI of these patients ranged from 30.9 to 62.7 kg/m², indicating that ORG is not limited to those with extreme obesity [121].

Research on renal pathology in individuals with severe obesity is limited. However, existing studies suggest early pathological changes even when significant clinical signs of renal disease are absent. Among 95 patients with severe obesity, with a mean BMI of 53.6 kg/m² who underwent renal biopsies before bariatric surgery, 41% showed microalbuminuria, while only 4 patients had albuminuria exceeding 300 mg/24 h. Glomerulomegaly was observed in 38% of cases, and focal segmental glomerulosclerosis in 5% [122].

These parameters have been studied across various ethnic groups, including Caucasian, Hispanic, and African American populations. The evaluation is complicated by factors like obesity, diabetes, and hypertension, which are interconnected and contribute to CKD progression. Choung et al. studied 248 morbidly obese subjects with a median BMI of 44, of whom 47% were diabetic and 81% hypertensive, with most having CKD. In 240 cases, detailed biopsy results showed ORG in 30% of the subjects, either alone or with other lesions. In the remaining 70%, findings included diabetic nephropathy, acute tubular necrosis, and hypertensive nephrosclerosis, with diabetic nephropathy being the most prevalent [123].

In summary, the diagnosis of obesity-related CKD requires a multifaceted approach combining traditional biomarkers, emerging urinary markers, and advanced imaging techniques. Novel molecular insights, particularly involving miRNAs, offer promising perspectives for early detection and intervention in obesity-related CKD. Furthermore, kidney biopsy studies emphasize the prevalence and characteristics of obesity-related CKD, suggesting that early pathological changes are commonly observed in these subjects. Continued research and development in these areas are essential for improving diagnostic accuracy and patient outcomes.

Renal Pathology in Obesity-Related CKD

Obesity-related glomerulopathy has become a notable cause of end-stage kidney disease, emerging as a primary renal condition in obese individuals. Significantly, ORG can develop independently of pre-existing renal diseases or overt diabetes, underscoring its importance as a distinct renal pathology [124]. Two primary hypotheses have been proposed to explain the glomerular abnormalities observed in ORG, encompassing hemodynamic alterations, and tubular feedback disruption [124]. In obese individuals, changes in hemodynamics lead to the dilation of glomerular inlet arteries, causing GH [125]. This hyperfiltration results in basement membrane expansion and glomerulomegaly. On the other hand, increased reabsorption of water and salt by the proximal tubule impairs the tubuloglomerular feedback mechanism [126, 127].

The key pathological features of ORG include oligonephronia, glomerulomegaly, and FSGS. Oligonephronia is characterized by a low density of glomeruli [124, 128]. Glomerulomegaly involves glomerular hypertrophy with mesangial proliferation, matrix accumulation, enlarged podocytes, and mild foot process fusion, while FSGS is defined by the segmental consolidation of the glomerular tuft by extracellular matrix and/or hyaline, resulting in capillary obliteration [129, 130]. ORG-associated FSGS typically progresses more slowly and mildly compared to primary FSGS, affecting fewer glomeruli and showing characteristic perihilar involvement due to increased filtration pressure at the afferent capillary network [131].

Podocytes, essential components of the glomerular filtration barrier, cover glomerular capillaries through interlaced foot processes [132]. In ORG, podocytes are subjected to mechanical strain from glomerular expansion, leading to their injury, dysfunction, and apoptosis. This process is critical for the development of proteinuria and the progression of ORG [133]. As podocytes cannot proliferate and have limited hypertrophic capacity, their eventual failure and detachment lead to localized denudation of the glomerular basement membrane and form a nidus for segmental sclerosis. With excessive weight gain, the single-nephron GFR increases, raising podocyte shear stress and the risk of podocytopathy [134]. To compensate for the reduced filtration area, remaining nephrons undergo hypertrophy, potentially leading to maladaptive podocyte hypertrophy. Hypertrophic podocytes may struggle to maintain normal foot process structure, resulting in increased local shear stress and subsequent podocyte separation [135].

Obesity-related glomerulopathy may also be complicated by renal interstitial fibrosis, tubular atrophy, and arteriosclerotic lesions, which may precede clinical evidence of renal impairment [124, 136]. Intracellular lipid vacuoles

may appear in proximal tubular epithelial cells, podocytes, and mesangial cells due to excessive lipid accumulation and deposition [137]. ORG may share histological features with diabetic nephropathy, even in the absence of glucose metabolism dysregulation. Approximately half of ORG patients exhibit ‘diabetoid’ changes, such as increased mesangial matrix, mild focal mesangial sclerosis, and mild focal thickening of glomerular and tubular basement membranes [138].

In ORG, immunofluorescence shows non-specific depositions of IgM and C3 in sclerotic and hyalinosis lesions, unlike immune-complex-type deposits, which are absent. Podocytes overlying sclerosis lesions may contain intracytoplasmic protein resorption droplets staining for IgG, IgA, and albumin [139]. Foot process effacement typically covers less than 50% of the glomerular surface area, in contrast to primary FSGS, where it exceeds 50% [140].

ORG represents a significant and distinct renal pathology associated with obesity, characterized by specific structural and functional changes in the kidneys. Understanding its pathophysiology and distinguishing it from similar conditions, particularly diabetic nephropathy is crucial for effective diagnosis and management.

Therapeutic Interventions

Lifestyle Interventions

Lifestyle modifications, involving a healthier diet and increased physical activity are a prerequisite for patients with obesity and CKD [141]. Mediterranean diet, a low-fat diet and a low carbohydrate diet have been compared in the Dietary Intervention Randomized Controlled Trial (DIRECT) study by Tirosh et al. In their study, Tirosh et al. have followed patients with obesity and CKD for a median of two years. They have concluded that the three dietary approaches compared had similar effects regarding amelioration in renal function as evidenced by increases in eGFR and reduction in albuminuria. These effects were more pronounced among patients with $\text{eGFR} < 60 \text{ mL/min/1.73m}^2$ and in patients with baseline microalbuminuria [142]. Regarding ketogenic diets the results are conflicting, yet. Very low-calorie ketogenic diet (VLCKD) is considered to be contraindicated in patients with CKD due to the increased risk for metabolic acidosis and nephrolithiasis [143, 144]. Nevertheless, Bruci et al., among patients with mild CKD, i.e. an eGFR between $60 \text{ mL/min/1.73m}^2$ and $89 \text{ mL/min/1.73m}^2$, reported that 27.7% exhibited an increase in eGFR to $\geq 90 \text{ mL/min/1.73m}^2$ after a VLCKD intervention [145]. Therefore, ketogenic diets in CKD should be rather plant-based and pursued under the strict supervision by a multidisciplinary team. Furthermore, current evidence suggests that they could be tried only if

other weight loss diets have failed and among patients with mild CKD [143–145].

Pharmacological Therapies

Antidiabetic Agents that Promote Weight Loss

Apart from lifestyle modifications, pharmacological treatment aiming at weight loss as well as improvement in renal function parameters is increasingly being used. Glucagon-like peptide-1 receptor (GLP-1) agonists and sodium glucose co-transporter 2 (SGLT-2) Inhibitors are two major classes of anti-diabetic drugs with well-recognized anti-obesity properties, especially GLP-1 receptor agonists. GLP-1 receptor agonists, such as liraglutide, semaglutide and dulaglutide, have been evaluated regarding weight loss in the LEADER, SUSTAIN 6 and REWIND studies, respectively. The aforementioned studies have demonstrated the beneficial outcomes of liraglutide, semaglutide and dulaglutide regarding weight loss and ameliorating renal function indices among patients with diabetes mellitus (DM) [146–149]. However, liraglutide 3 mg/daily has already been associated with weight loss among patients with overweight/obesity without DM [150]. Besides, semaglutide at a dose of 2.4 mg once weekly has been shown to have beneficial effects among patients with overweight/obesity and without diabetes mellitus in the STEP 1 clinical trial [151].

In sharp contrast, SGLT-2 Inhibitors have not gained FDA approval for the treatment of obesity. However, due to their unique mechanism of action, which includes increasing renal glucose excretion and thereby causing a net energy loss, SGLT2 inhibitors could be promising candidates for treating obesity [152–154]. It is noteworthy that both classes of drugs, GLP-1 agonists and SGLT-2 inhibitors, have proven beneficial effects on preventing further renal function decline. More specifically, empagliflozin and canagliflozin in the EMPAREG renal outcomes and the CREDENCE clinical trials respectively, have demonstrated amelioration in renal function indices among patients with CKD and DM. The DAPA-CKD clinical trial has documented substantial improvement in renal dysfunction among patients with CKD, irrespective of the presence or absence of DM [155–157].

In the clinical spectrum of incretins, tirzepatide is the first dual GLP-1/GIP (glucose-dependent insulintropic peptide) agonist. Tirzepatide has been developed based on the concept of combining these two incretins in one co-agonist in order to augment the beneficial effects of each incretin in one molecule [158]. Indeed, tirzepatide has been documented to reduce body weight more than semaglutide among patients with DM in the SURPASS-2 clinical trial (-5.5 kg relative difference) [159]. Regarding weight loss among patients

with obesity only, tirzepatide has been shown to promote a reduction in body weight of 21% at the highest dose of 15 mg once weekly in the SURMOUNT-1 clinical trial [160]. Therefore, tirzepatide promotes further weight loss, above and beyond GLP-1 receptor agonists and with no more statistically significant adverse effects from the gastrointestinal tract [160].

Other Drugs

Apart from weight loss drugs, molecules that act via RAAS inhibition, such as ACE Inhibitors, ARBs and mineralocorticoid receptor antagonists (MRAs) or aldosterone antagonists are very useful in preserving renal function. MRAs are classified as steroidal, such as spironolactone and eplerenone and non-steroidal, such as finerenone. Although neutral regarding body weight, these molecules have been documented to improve renal dysfunction. Lately, the combination of ACEI or ARBs with a MRA, especially finerenone, is being extensively studied, even among patients without DM [8, 161].

Finally, melatonin, which is a hormone secreted from the pineal gland, has been suggested to improve obesity-related CKD. Melatonin plays a pivotal role in regulating circadian rhythms. Melatonin is mainly released at night and exerts its biological effects, i.e. regulation of the biological clock, via its receptors, which are widespread throughout the human body. In the kidney, melatonin exhibits anti-inflammatory properties by inactivating the NLRP3 inflammasome [162]. Besides, it acts as a ROS scavenger, thereby reducing oxidative stress and inflammation as well [163]. In addition, its anti-obesity effects have been related to its ability to increase the amount of brown adipose tissue (BAT) and thus, energy expenditure [164]. The aforementioned mechanisms of action of melatonin in obesity-related CKD advocate its utility in ameliorating renal dysfunction associated with obesity. Table 1 presents clinical study evidence linking obesity-related CKD with potential drug therapies.

Bariatric Surgery

Bariatric surgery has the best results in terms of weight loss until now, mainly due to alterations in gastric hormones and the subsequent mitigation in inflammation and insulin resistance [165, 166]. However, tirzepatide's high dose once weekly has been reported to confer a 21% reduction in body weight, which is quite similar to those of bariatric surgery [157]. These remarkable results of the SURMOUNT-1 clinical trial may shed light on further therapeutic approaches regarding weight loss interventions. Taking into consideration various side effects of bariatric surgery, such as the increased risk of deep venous thrombosis and pulmonary embolism, as well as gastrointestinal and infectious adverse

effects, tirzepatide perhaps will shed light on future research regarding weight loss techniques [167]. Notably, a triple G receptor agonist retatrutide, i.e. GLP-1/GIP/GCG (glucagon) is just at the dawn of research [168]. It has been reported that retatrutide resulted in a 24% mean weight loss at 48 months [168]. It seems likely that we will enrich our armamentarium in the combat against obesity in the near future [169]. Table 2 depicts on-going studies in humans regarding obesity-related CKD and potential treatment options.

Future Perspectives and Controversies

According to WHO, in 2022, it has been estimated that 1 in 8 people worldwide were affected by obesity [WHO, Obesity facts, assessed 26th June 2024], which is associated with an array of associated disorders [86, 170–179]. CKD in the context of obesity poses a challenge for clinicians. Weight loss lies at the heart of the problem. Lifestyle interventions and drugs, such as GLP-1 receptor agonists, and lately dual GLP-1/GIP agonists or triple G receptor agonists seem to be very promising. However, dual GLP-1/GIP and triple G receptor agonists are relatively novel drugs that should be further evaluated in large scale studies in order to assess their effectiveness and safety. In addition, dual and triple G receptor agonists should be re-evaluated compared to bariatric surgery regarding their weight loss potential in the near future. Moreover, GLP-1 receptor agonists, dual and triple G receptor agonists should be assessed regarding their nephroprotective properties as well.

Drugs that act via RAAS inhibition have been extensively used in the treatment of CKD of any cause. However, further studies are needed in order to examine the utility of steroidal MRAs together with non-steroidal MRAs, such as finerenone in obesity-related CKD. Finally, melatonin supplementation in the context of CKD associated with obesity should be evaluated in future studies. Nowadays, the perpetual discovery of novel drugs aiming at combating obesity and obesity-related CKD seems to be a fact not a fiction. It remains to be elucidated to what extent these drugs and their combinations ameliorate CKD associated with obesity.

Numerous equations have been proposed to estimate GFR. At present, international societies recommend the CKD Epidemiology Collaboration (CKD-EPI) equation [180–184]. Nevertheless, the CKD-EPI equation has several significant limitations: it has not been validated in individuals with obesity and assumes a body surface area (BSA) of 1.73 m², which reflects the average BSA of 25-year-old Americans from a century ago [185, 186]. This figure is considerably smaller than the average BSA of today's population [186, 187]. As a result, indexing GFR to this outdated BSA underestimates renal function in about one-third of the

population, leading to an artificially higher prevalence of CKD. Conversely, this indexing decreases the true prevalence of glomerular hyperfiltration, a still neglected subgroup with increased kidney and cardiovascular risk [186]. Directly measured GFR is considered the gold standard; however, its high cost and relative impracticality prevent its use in routine practice [188]. Using the CKD-EPI equation in subjects with obesity results in an overestimation of CKD prevalence and an underestimation of GHF prevalence. De-indexed equations that account for individual BSA should be recommended for epidemiological studies until direct GFR measurement becomes more accessible [186].

While obesity, defined by a BMI > 30 kg/m², especially in the context of metabolic syndrome and insulin resistance, constitutes a risk factor for de novo CKD, with a rapid loss of kidney function in patients with eGFR > 60 ml/min/1.73 m² [189], it may confer survival advantages in advanced CKD, the so-called obesity paradox [190–192]. Meta-analyses have indicated that once CKD develops, higher BMI ranges (overweight and obese) are paradoxically linked to better survival rates in advanced predialysis (eGFR < 30 ml/min/1.73 m²) and dialysis-dependent CKD patients [193]. However, a pooled analysis revealed that higher pre-transplantation BMI is associated with increased mortality in kidney transplant recipients [194]. Besides advanced CKD, the obesity paradox has also been noted in heart failure, chronic obstructive lung disease, liver cirrhosis, metastatic cancer, and among elderly individuals [195–201]. In these disease states, protein energy wasting, and inflammation are important predictors of early death. Greater body adipose tissue and, in particular, increased muscle mass may contribute to longer survival in these patients. The obesity paradox may stem from a timing discrepancy between competing risk factors: overnutrition, which is a long-term threat, versus undernutrition, a short-term threat. Parameters including the hemodynamic stability provided by obesity, the lipoprotein defense against circulating endotoxins, the protective and anti-inflammatory cytokine patterns, the ability of the adipose tissue to sequester toxins, and the anti-oxidative, anti-inflammatory and beneficial metabolic properties of muscle tissue may play significant roles [190, 202–205]. Nevertheless, this obesity paradox may result from residual confounding. Investigating the causes and implications of the obesity paradox in patients with advanced CKD could help elucidate key pathogenetic mechanisms, resulting in better outcomes for subjects with CKD.

Table 2 List of ongoing studies in humans associating Obesity-related CKD and potential drug therapeutics

Title of Study/NCT	Human Study Characteristics	Places of the Study/Information	Status of the Study
<i>List of Studies with GLP-1 receptor agonists, dual GLP-1/GIP and triple G receptor agonists</i>			
Semaglutide and Albuminuria Reduction Trial in Obese Individuals Without Diabetes (SMART) [NCT04889183]	Age $\geq$ 18 years-old BMI $\geq$ 27 kg/m ² eGFR $\geq$ 25 ml/min/1.73m ² No DM 125 participants Phase 3 Study, which started 2022/03/14 and finished 2024/05/28	Conducted in Canada, Germany, Netherlands, Spain Information provided by University Medical Center Groningen	- Completed - Awaiting Results with Semaglutide up to 2.4 mg
Pharmacokinetics, Safety and Tolerability of Oral Semaglutide in Subjects With Various Degrees of Impaired Renal Function [NCT02014259]	Age $\geq$ 18 years old 71 Participants Phase 1 Study, which started 2013/12/11 and finished 2014/10/24	Conducted in Czechia and Hungary Information provided by Novo Nordisk	- Completed - No effect on the dose of oral semaglutide up to 10 mg among patients with CKD, even ESRD
A Study of Tirzepatide (LY3298176) in Participants With Impaired Kidney Function [NCT03482024]	Ages between 18 and 85 years old BMI between 19 and 40 kg/m ² 45 Participants Phase 1 Study, which started 2018/03/30 and finished 2019/08/19	Conducted in the U.S.A Information provided by Eli Lilly and Company	- Completed - Tirzepatide at a dose of 5 mg could be administered safely among patients with CKD, even ESRD
The Effect of Retatrutide Once Weekly on Cardiovascular Outcomes and Renal Function in Adults Living With Obesity (TRIUMPH-OUTCOMES) [NCT06383390]	Age $\geq$ 45 years old BMI $\geq$ 27 kg/m ² CKD With or without DM Phase 3 Study, which started 2024/04/30 and is still recruiting about an estimated 10,000 participants	Conducted in the U.S.A, Argentina, Australia, Austria, Belgium, Brazil, Canada, Czechia, Denmark, France, Germany, Greece, Hungary, India, Israel, Italy, Mexico, Netherlands, Poland, Puerto Rico, Romania, Slovakia, Spain, Turkey, United Kingdom Information provided by Eli Lilly and Company	Still Recruiting
A Study of Retatrutide (LY3437943) on Renal Function in Participants With Overweight or Obesity and Chronic Kidney Disease With or Without Type 2 Diabetes [NCT05936151]	Age $\geq$ 18 years old BMI $\geq$ 27 kg/m ² CKD With or without DM Estimated 120 Participants Phase 2 Study, which started 2023/07/20 and is still recruiting	Conducted in the U.S.A., Canada, Italy, Spain, United Kingdom Information provided by Eli Lilly and Company	Still Recruiting
<i>List of Studies with combination of SGLT-2 Inhibitors and GLP-1 Analogs</i>			
Comparison of the Efficacy and Safety of SGLT2i and GLP-1 Receptor Agonists in Obese Patients With Kidney Disease [NCT06344247]	Age between 18 and 75 years old BMI $\geq$ 25 kg/m ² or WC $\geq$ 85 cm for male and $\geq$ 80 cm for female Obesity-related CKD confirmed by biopsy On optimized therapy with ACEI/ARBs and/or MRAs for the last 3 months, minimum Estimated 48 Participants Phase ? Study, which started 2023/09/01 and is still recruiting	Conducted in China Information provided by Renji Hospital in China	Still Recruiting

Abbreviations: ACEI: Angiotensin Converting Enzyme Inhibitor; ARBs: Angiotensin Receptor Blockers; BMI: Body Mass Index; CKD: Chronic Kidney Disease; DM: Diabetes Mellitus; eGFR: estimated Glomerular Filtration Rate; MRAs: Mineralocorticoid Receptor Antagonists; NCT: Number of Controlled Trial; WC: Waist Circumference

Conclusions

Obesity significantly contributes to the onset and progression of CKD, underscoring the need for targeted public health strategies. Understanding the complex interactions between obesity, chronic inflammation, oxidative stress, and insulin resistance is crucial for developing effective interventions. The role of RAAS in obesity-related hypertension and CKD progression highlights promising therapeutic perspectives. Furthermore, the relationship between gut microbiota and kidney health emphasizes the importance of maintaining gut integrity to prevent CKD [87, 206, 207]. Advances in diagnostic techniques, including molecular markers and imaging, are vital for early detection and management [208]. Ongoing research into obesity-related renal pathology will enhance diagnostic precision and treatment efficacy, ultimately improving patient outcomes.

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169. Patti ME. Triple G Agonists—A Home Run for Obesity? *N Engl J Med*. 2023;389(6):562–563. doi: 10.1056/NEJMe2307282.

In this Editorial, the potential of Triple G Agonists is further elucidated.

Abbreviations ACE: Angiotensin Converting Enzyme; ACEI: Angiotensin Converting Enzyme Inhibitor; ADIPOR: Adiponectin Receptor; ApoL1: Apolipoprotein L1; ARBs: Angiotensin Receptor Blockers; ASM: Acid Sphingomyelinase; ATF6: Activating Transcription Factor 6; ATMs: Adipose Tissue Macrophages; BCFAs: Branched Chain Fatty Acids; BMI: Body Mass Index; CC: Chemokines; CKD: Chronic Kidney Disease; CKD-EPI: CKD Epidemiology Collaboration; CT: Computed Tomography; DAG: Diacylglycerol; ECM: Extracellular Matrix; eGFR: Estimated Glomerular Filtration Rate; EMT: Epithelial-Mesenchymal Transition; ER: Endoplasmic Reticulum; ERPF: Effective Renal Plasma Flow; ESRD: End-stage Renal Disease; FFAs: Free Fatty Acids; FSGS: Focal Segmental Glomerulosclerosis; GCG: Glucagon; GFR: Glomerular Filtration Rate; GH: Glomerular Hyperfiltration; GHF: Glomerular hyperfiltration; GIP: Glucose-dependent Insulinotropic Peptide; GLP-1: Glucagon Like Peptide-1; HIF-1 α : Hypoxia-Inducible Factor 1 α ; ICAM-1: Inter-cellular Adhesion Molecule 1; IL: Interleukin; IL-1Ra: Interleukin-1 Receptor Antagonist; IRE1: Inositol Requiring Enzyme 1; IRS-1: Insulin Receptor Substrate-1; IS: Indoxyl Sulfate; JAK2: Janus Kinase 2; JNK: C-Jun N-terminal Kinase; LPS: Lipopolysaccharide; MCP-1: Monocytes Chemoattractant Protein-1; MHO: Metabolically Healthy Obese; miRNAs: MicroRNAs; MMP: Matrix Metalloproteinase; MRAs: Mineralocorticoid Receptor Antagonists; MRI: Magnetic Resonance Imaging; NF- κ B: Nuclear Factor kappa-light-chain-enhancer of Activated B cells; NHANES: National Health and Nutrition Examination Survey; NLRP3: Nucleotide binding oligomerization Domain Like Receptor Protein 3; NGS: Next Generation Sequencing; ORG: Obesity-Related Glomerulopathy; PERK: Protein Kinase RNA like ER Kinase; PPAR: Peroxisome Proliferator Activated Receptor; PAI-1: Plasminogen Activator Inhibitor-1; PCR: Polymerase Chain Reaction; PCS: P Cresyl Sulfate; PKC: Protein Kinase C; RAAS: Renin Angiotensin Aldosterone System; ROS: Reactive Oxygen Species; RPF: Renal Plasma Flow; SCFAs: Short Chain Fatty Acids; SCr: Serum Creatinine; sCysC: Cystatin C; SGLT-2: Sodium Glucose Cotransporter-2; STAT3: Signal Transducer and Activator of Transcription 3; TREM2: Triggering Receptor Expressed on Myeloid Cells 2; TGF- β : Transforming Growth Factor β ; TJs: Tight Junctions; TMA: Trimethylamine; TNF- α : Tumor Necrosis Factor alpha; UPR: Unfolded Protein Response; VCAM-1: Vascular Cell Adhesion Molecule 1; VLCKD: Very Low Calorie Ketogenic Diet; WAT: White Adipose Tissue; WHO: World Health Organization

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performed literature research; DK, NV, MV and IK prepared the tables. DK prepared the figures; DK, NV and MV reviewed the manuscript.

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Compliance with Ethical Standards

Competing Interests The authors declare no competing interests.

Human and Animal Rights and Informed Consent This article does not contain any studies with human or animal subjects performed by any of the authors.

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